

<Library Management System> Final Report

SEMESTER MARCH-AUGUST 2024

GROUP: CS110 4A

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<Library Management System>	Version: <1.0>
Final Report	Date: <07/07/2024>
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Revision History

Date	Version	Description	Author
<08/07/2024>	<1.0>	<Update new system>	<Mohammad Amarulhariz Bin Mohd Fairuz>

<Library Management System>	Version: <1.0>
Final Report	Date: <07/07/2024>
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1. Introduction

1.1 Purpose

This final report's objective is to chronicle and assess the library system project's entire journey. Planning is the first step, during which project objectives and preliminary requirements are established. Following extensive system analysis to hone requirements and look into solutions, fact-finding activities collected information about user needs and current systems. The architecture, user interfaces, and database structure were then influenced by system design. The process of implementation included building, testing, and deploying the system. In summary, the report evaluates the system's performance in achieving its objectives, shares insights gained, and makes recommendations for future development and user satisfaction.

1.2 References

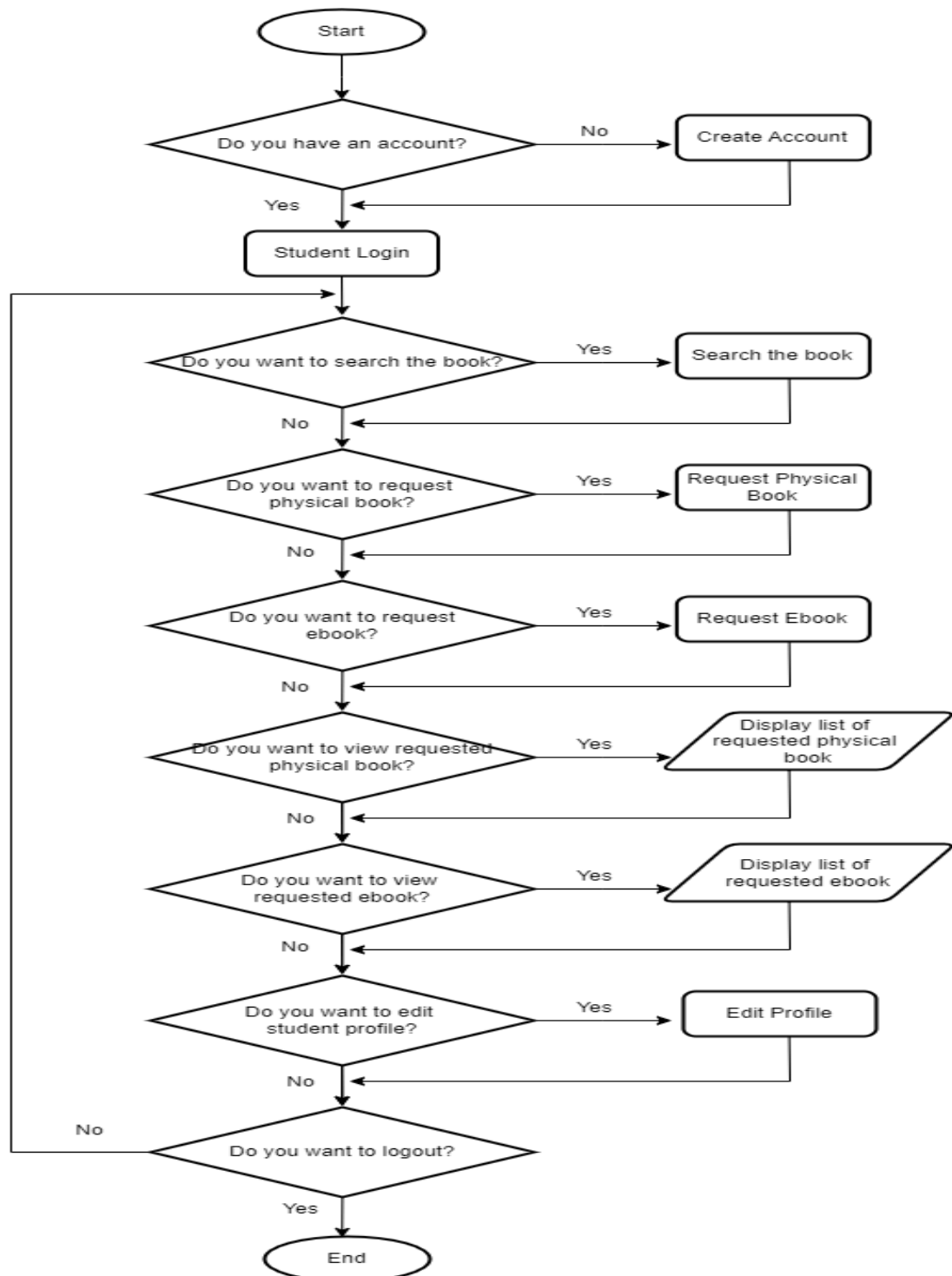
1. Shelly, G. B., & Rosenblatt, H. J. (2011, January 1). *Systems Analysis and Design*. http://books.google.ie/books?id=NP1UmgEACAAJ&dq=System+analysis+and+design+9th+edition&hl=&cd=6&source=gbs_api
2. Using ajax for every div with class/id. (n.d.). Stack Overflow. <https://stackoverflow.com/questions/48335498/using-ajax-for-every-div-with-class-id>
3. Ranjan, A., Sinha, A., & Battewad, R. (2020). *JavaScript for modern web development: building a web application using HTML, CSS, and JavaScript*. BPB Publications.

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2. Project Overview

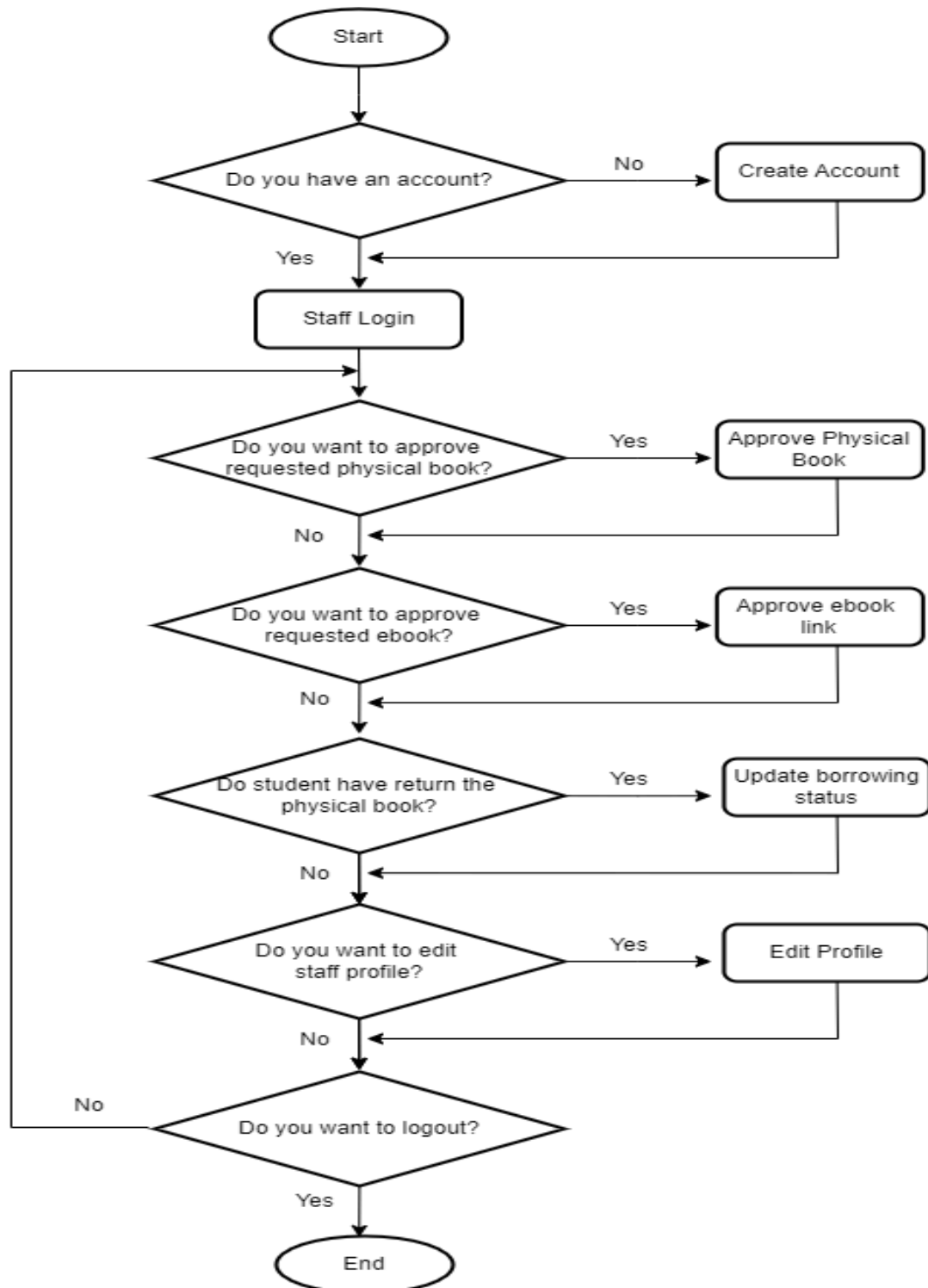
2.1 Project System Process

Student Flowchart



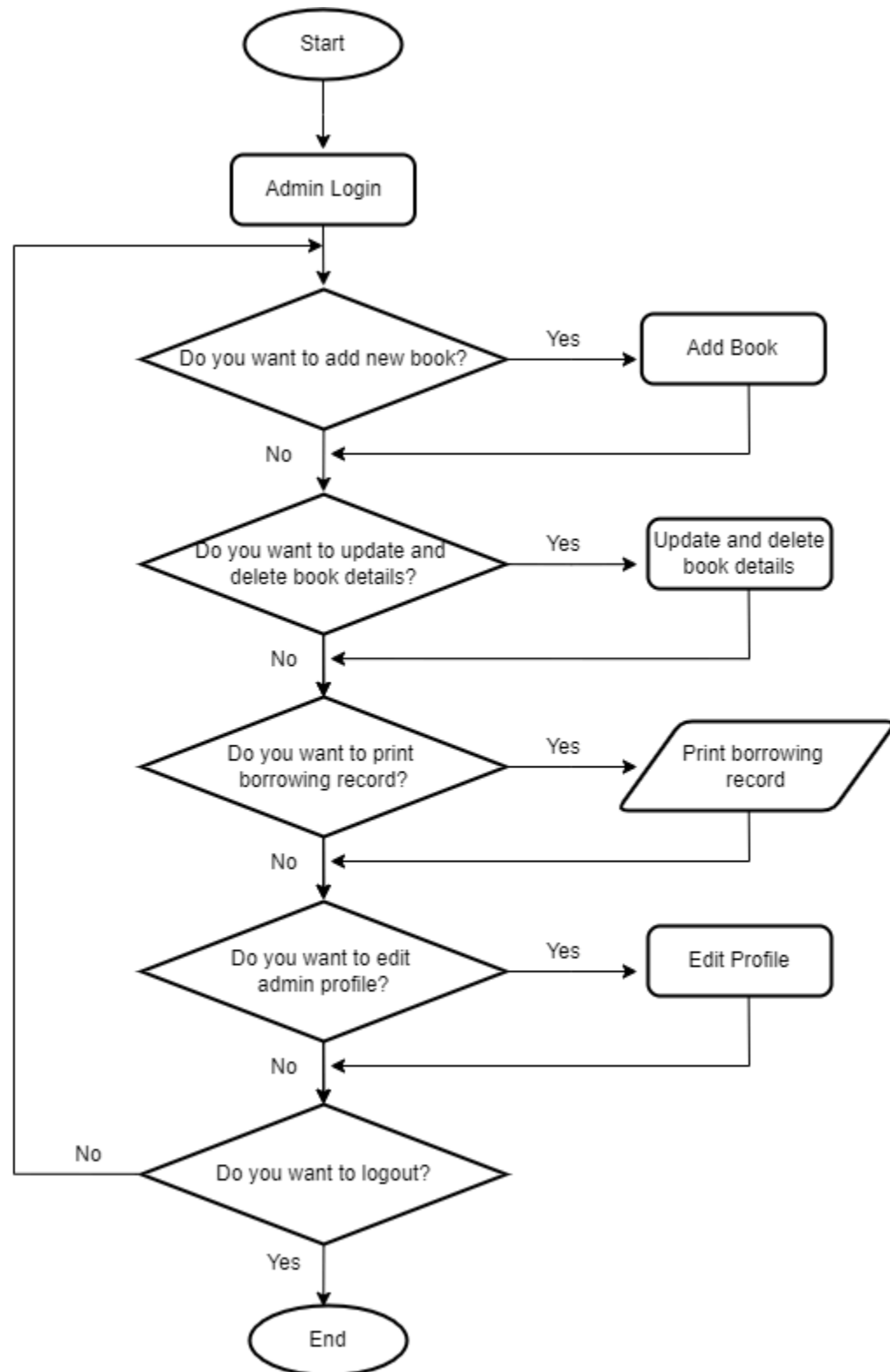
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Staff Flowchart



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Admin Flowchart



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2.2 Project scope

Student : Have the access to search books, request books such as physical books and ebooks and also view a list of requested physical books and ebooks.

Staff : Have the access to approve or reject requested books by students and update borrowing status when students return the physical book.

Admin: Have the access to update and delete book details, add books and generate borrowing records.

Limitation of the system

- **Fine Management**: Managing fines for overdue books, including calculation, payment tracking, and fine forgiveness policies.
- **Reservation System**: Allowing users to reserve books that are currently checked out, with notifications when reserved books become available.
- **User Ratings and Reviews**: Allowing users to rate and review books in the catalog.

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2.3 Project objectives

1. To analyze the requirements for the new system.

To ensure that the new Library Management System satisfies stakeholders' expectations and enhances present operations, conduct a thorough investigation to determine their needs and requirements. This entails interviews from administrators and staff to obtain specific requirements and pinpoint issues with the current system. Both functional and non-functional needs will be included in the documentation of the information acquired, and a gap analysis will be carried out to determine what features and capabilities are currently missing from the system and to identify any shortcomings. Prioritizing the requirements will ensure that the most important aspects are taken care of first and that they comply with academic, data protection, and regulatory standards. The requirements will be ranked in order of priority and viability.

2. To design an Intuitive user interface for library management systems.

Develop a user-friendly and intuitive interface that enhances the user experience for all stakeholders, making the system easy to navigate and use. Iterative usability testing with real users will refine the interface design, ensuring it meets usability standards and complies with accessibility standards. A consistent design language and style guide will be developed to maintain uniformity across the system's interface.

3. To develop a web based library management system for UiTM.

Build a robust, scalable, and secure web-based LMS that integrates seamlessly with existing systems and provides a comprehensive solution for library management at UiTM. Core Library Management System features like catalog management, user account management, and digital resource access will be implemented with robust security measures to protect user data and ensure compliance with data protection regulations. Through testing, it will ensure the system is reliable and bug-free, and comprehensive documentation and training materials will be provided to ensure smooth adoption and ongoing use of the system.

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2.4 Assumptions and constraints

1. Device Compatibility

The Library Management System might not be compatible with all devices, particularly older hardware with limited processing power, memory, or outdated operating systems. This can restrict students and staff from accessing the system on their personal devices.

2. Data Security

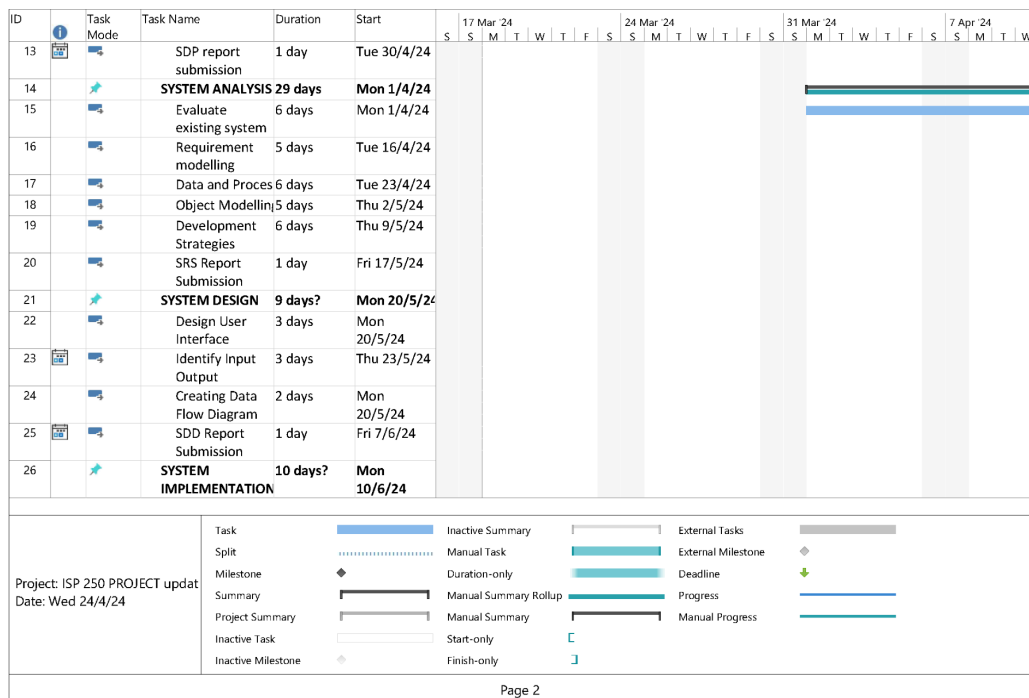
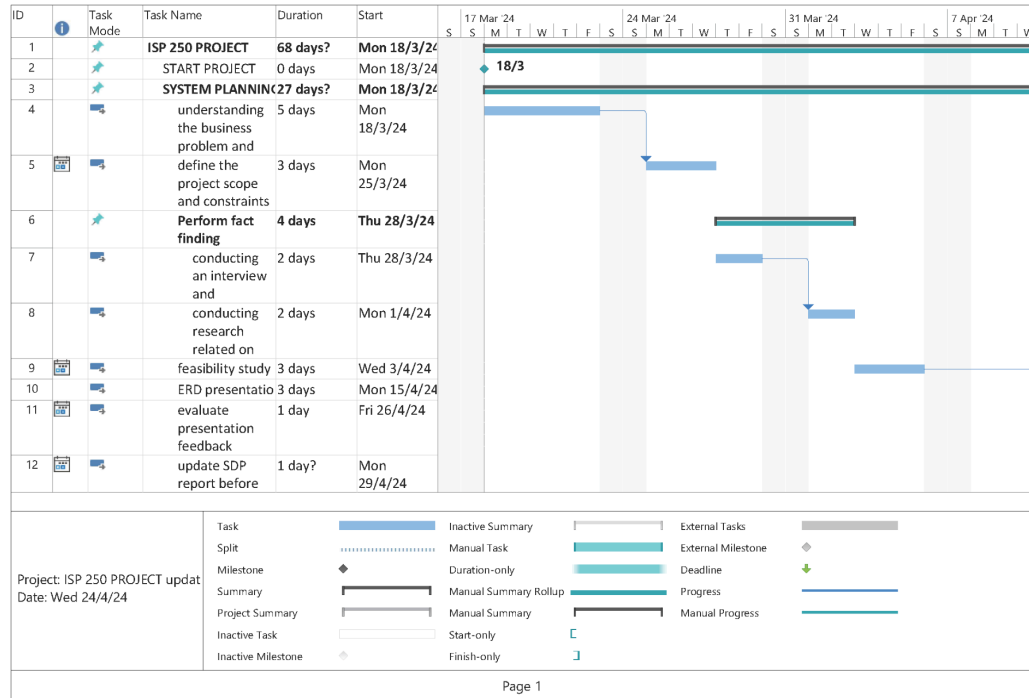
Ensuring the security of sensitive user data, such as personal information and borrowing histories, is paramount. Limitations in implementing advanced security measures can expose the system to risks like data breaches and unauthorized access.

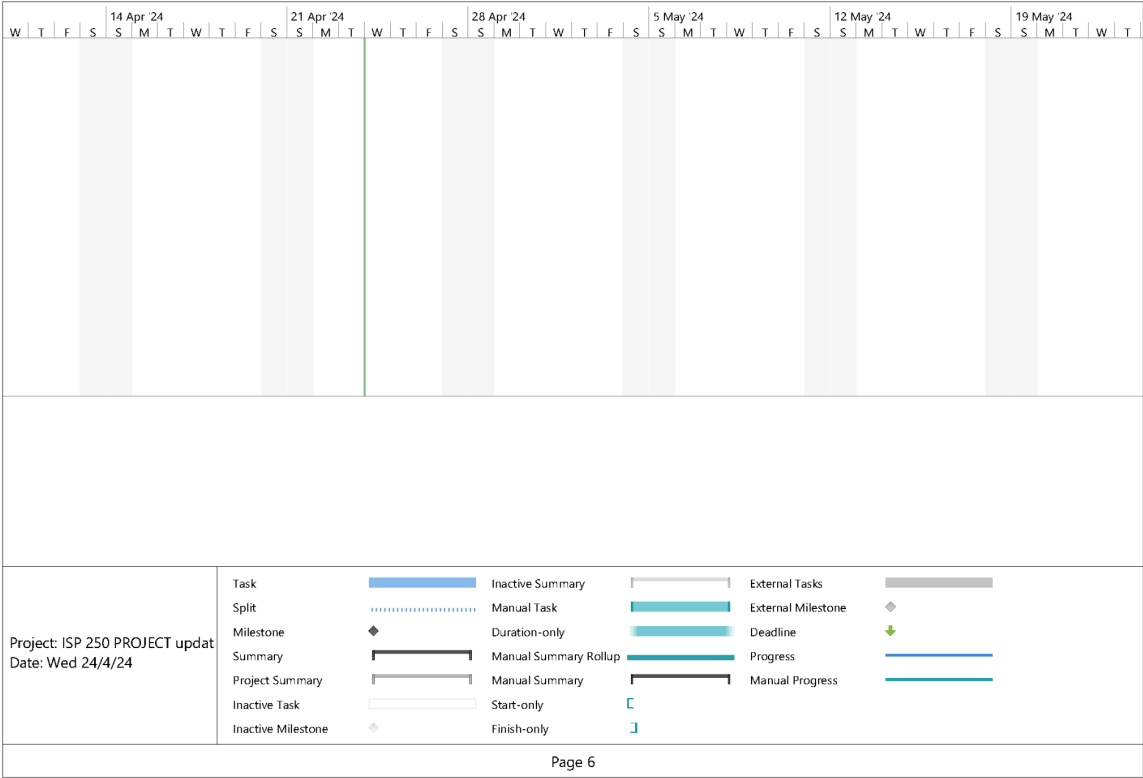
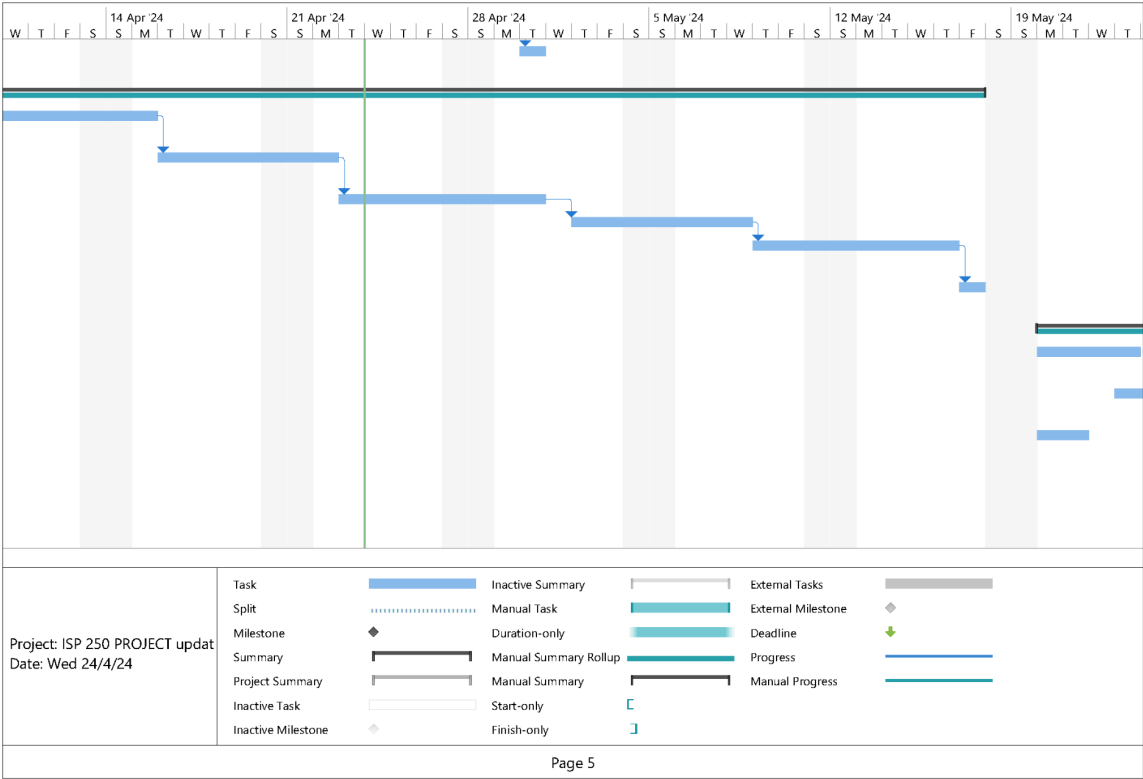
3. Network Dependence

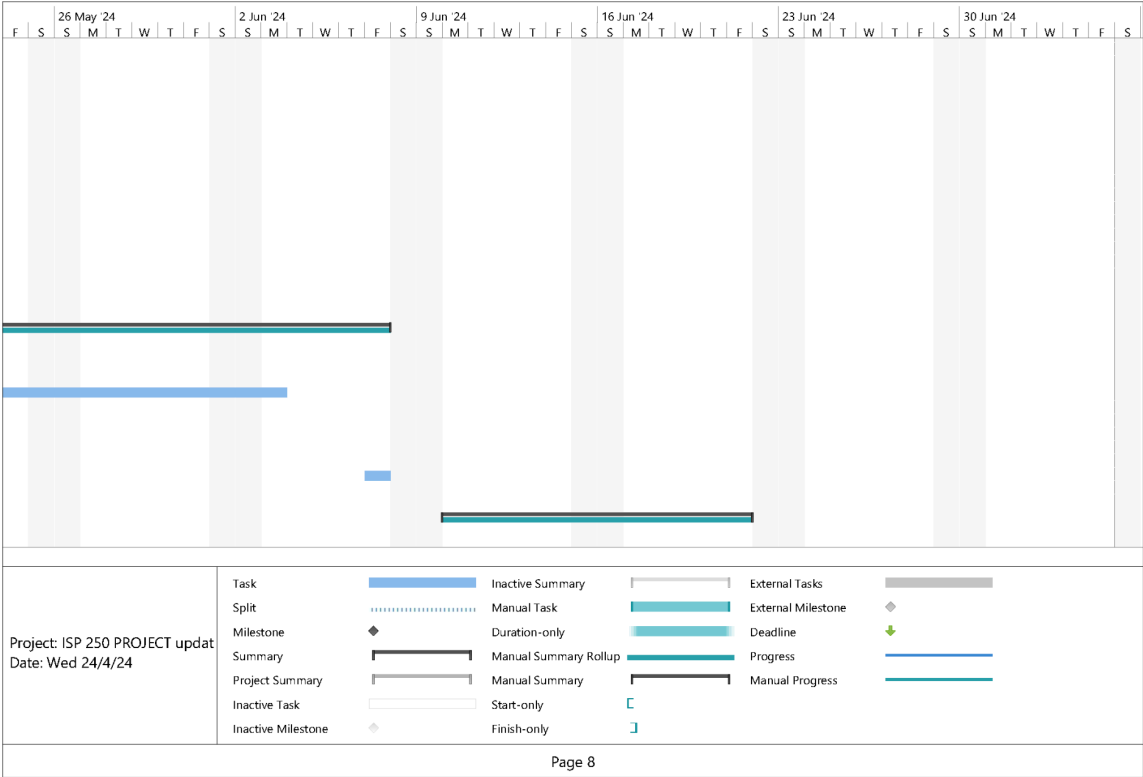
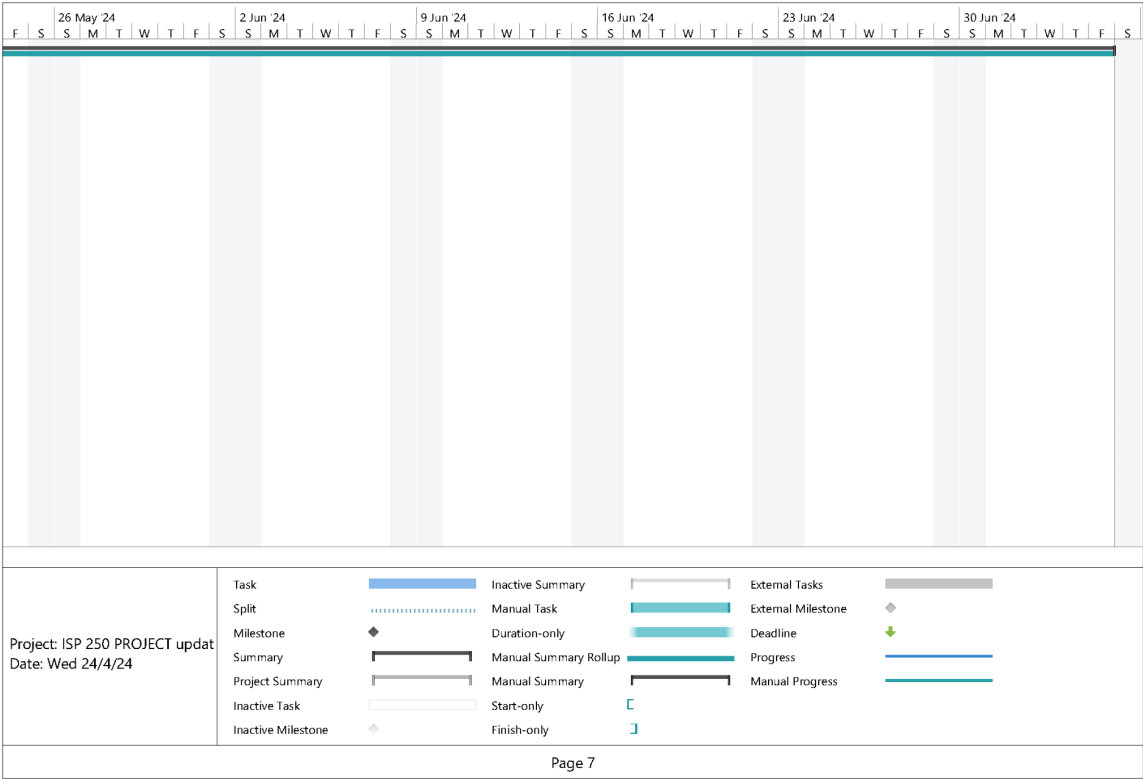
The Library Management System relies heavily on network connectivity. Poor internet connectivity or network outages can significantly impact the accessibility and performance of the system for all stakeholders.

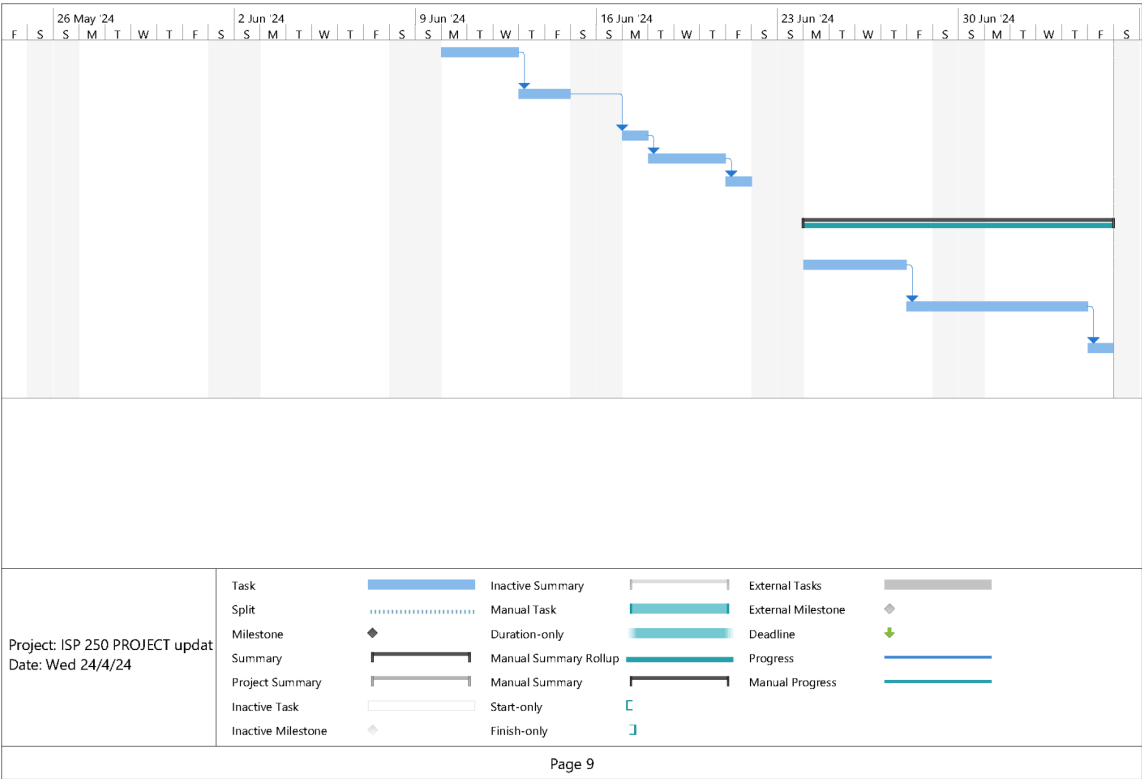
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2.5 Project deliverables









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3. Project Organization

3.1 Organizational Structure



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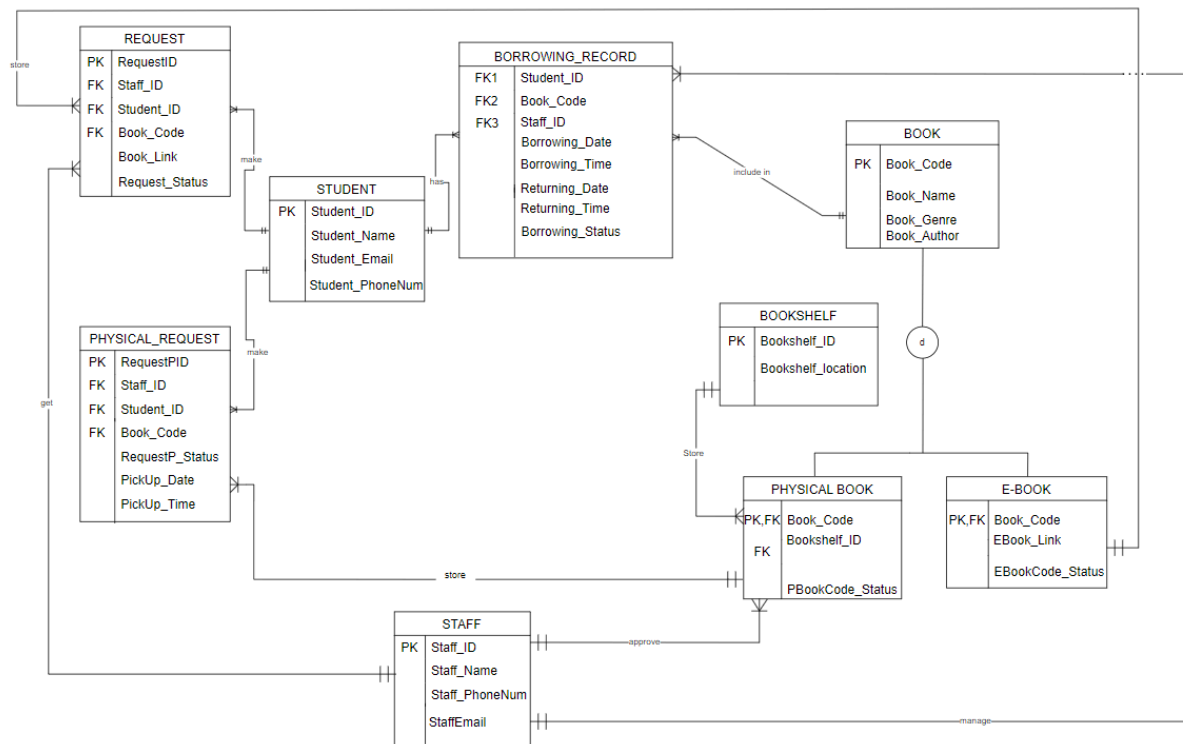
3.2 Roles and Responsibilities

Members	Role	Responsibilities
 Mohammad Amarulhariz Bin Mohd Fairuz	Project Manager	• Planning project activity • Organize the project • Monitor project progress
 Muhammad Aiman Bin Mohd Nazri	System Analyst	• Analyzing organization's current computer systems • Identify problems within the system • Propose solutions for the problems
 Muhammad Haiqal Bin Saharudin	Database Designer	• Design database schemas and create databases • Track the performance of the databases
 Muhammad Syazwi Hilmi Bin Saifol Bakhri	Programmer	• Writing code • Testing and debugging • Problem solving • Research

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4. Database and System Design

4.1 Entity Relationship Diagram



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4.2 Data Dictionary

Table Name	Attribute Name	Content	Type	Format	Range	Required	PK or FK	FK References Table
Student	Student_ID	Student ID	int (10)	123456789		Yes	PK	
	Student_Name	Student Full Name	VarChar(40)	Xxxxxxx123				
	Student_Email	Student Email	VarChar(100)	Xxxxxxx123				
	Student_PhoneNum	Student Phone Number	VarChar(128)	Xxxxxxx123				
Staff	Staff_ID	Staff ID	int(12)	123456789		Yes	PK	
	Staff_Name	Staff Full Name	VarChar(40)	Xxxxxxx123				
	Staff_Email	Staff Email	VarChar(100)	Xxxxxxx123				
	Staff_PhoneNum	Staff Phone Number	int(10)	123456789				
Request	Request_ID	Request ID of ebook	int(100)	123456789		Yes	PK	
	Student_ID	Student ID	int (10)	123456789		Yes	FK	Student

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	Staff_ID	Staff ID	int(12)	123456789		Yes	FK	Staff
	Book_Code	Book Code	VarChar(10)	Xxxxxxx123		Yes	FK	Book
	EBook_Link	Book Link	Varchar(128)	Xxxxxxx123				
	Request_Status	Request Status of E-Book	Varchar(12)	Xxxxxxx123				
Physical Request	RequestP_ID	Request ID of Physical Book	int(100)	123456789		Yes	PK	
	Student_ID	Student ID	int (10)	123456789		Yes	FK	Student
	Staff_ID	Staff ID	int(12)	123456789		Yes	FK	Staff
	Book_Code	Book Code	VarChar(10)	Xxxxxxx123		Yes	FK	Book
	RequestPhysical_Status	Request Status of Physical Book	VarChar(12)	Xxxxxxx123				
	PickUp_Date	Date to Pick Up the book	Date	YY-MM-DD				
	PickUp_Time	Time to Pick Up the book	Time	00-00-00				
Book	Book_Code	Book Code	VarChar(10)	Xxxxxxx123		Yes	PK	
	Book_Name	Name of the book	VarChar(40)	Xxxxxxx123				
	Book_Author	Author Name of the book	VarChar(40)	Xxxxxxx123				
	Book_Genre	Type of book	VarChar(40)	Xxxxxxx123				
E-Book	Book_Code	Book Code	VarChar(10)	Xxxxxxx123		Yes	PK/FK	Book
	EBook_Link	Link of E-book	VarChar(128)	Xxxxxxx123				
	EBookCode_Status	Status of E-Book	VarChar(12)	Xxxxxxx123				
Physical Book	Book_Code	Book Code	VarChar(10)	Xxxxxxx123		Yes	PK/FK	Book
	BookShelf_ID	Location of the Book	VarChar(10)	Xxxxxxx123		Yes	FK	BookShelf
	Physical_Status	Status of Physical Book	VarChar(12)	Xxxxxxx123				

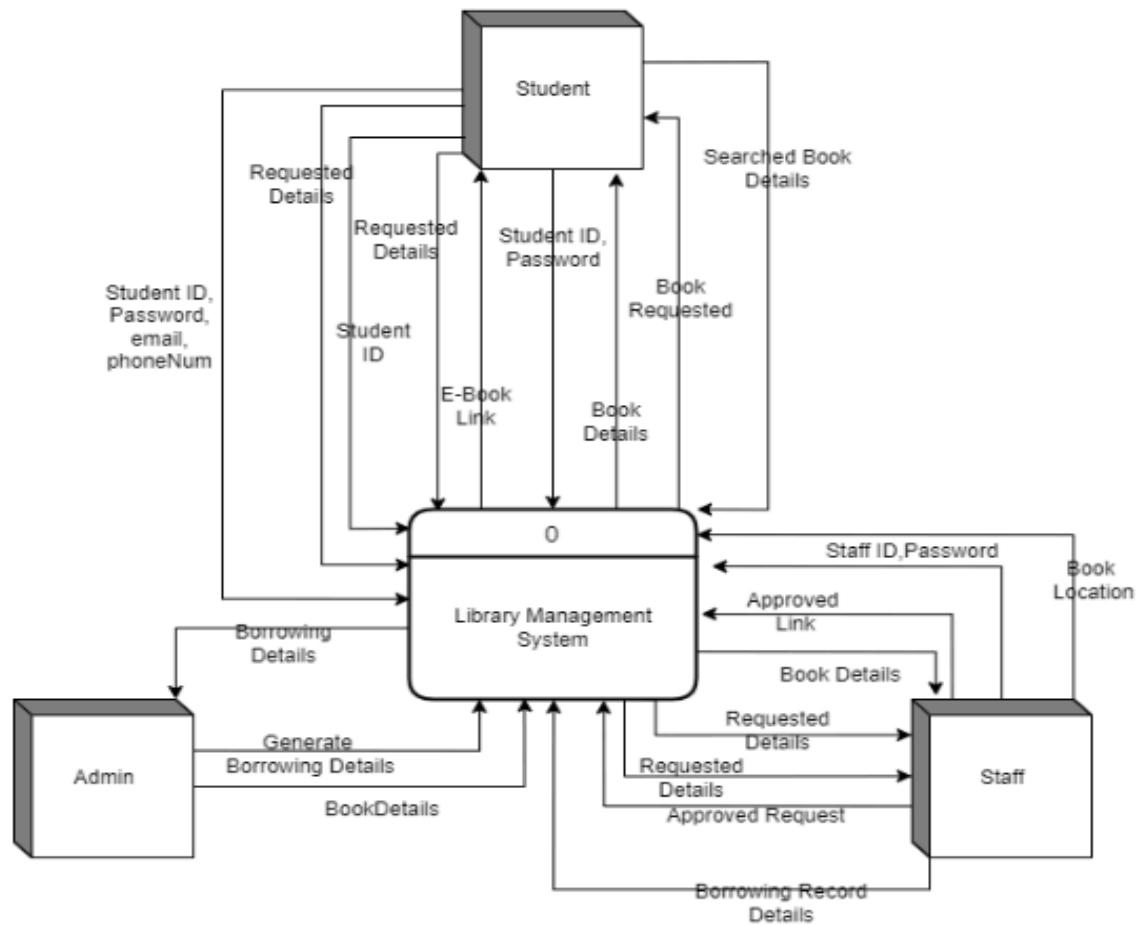
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Borrowing _Record	Student_ID	Student ID	int (10)	123456789		Yes	FK	Student
	Staff_ID	Staff ID	int(12)	123456789		Yes	FK	Staff
	Book_Code	Book Code	VarChar(10)	Xxxxxxx123		Yes	FK	Book
	Borrowing_Date	Date of borrowing	Date	YY-MM-DD				
	Borrowing_Time	Time of Borrowing	Time	00-00-00				
	Returning_Date	Date of Returning	Date	YY-MM-DD				
	Returning_Time	Time of Returning	Time	00-00-00				
	Borrowing_Status	Status of borrowing	VarChar(12)	Xxxxxxx123				
BookShelf	BookShelf_ID	Location ID	VarChar(10)	Xxxxxxx123		Yes	PK	
	BookShelf_Location	Location level	VarChar(10)	Xxxxxxx123				

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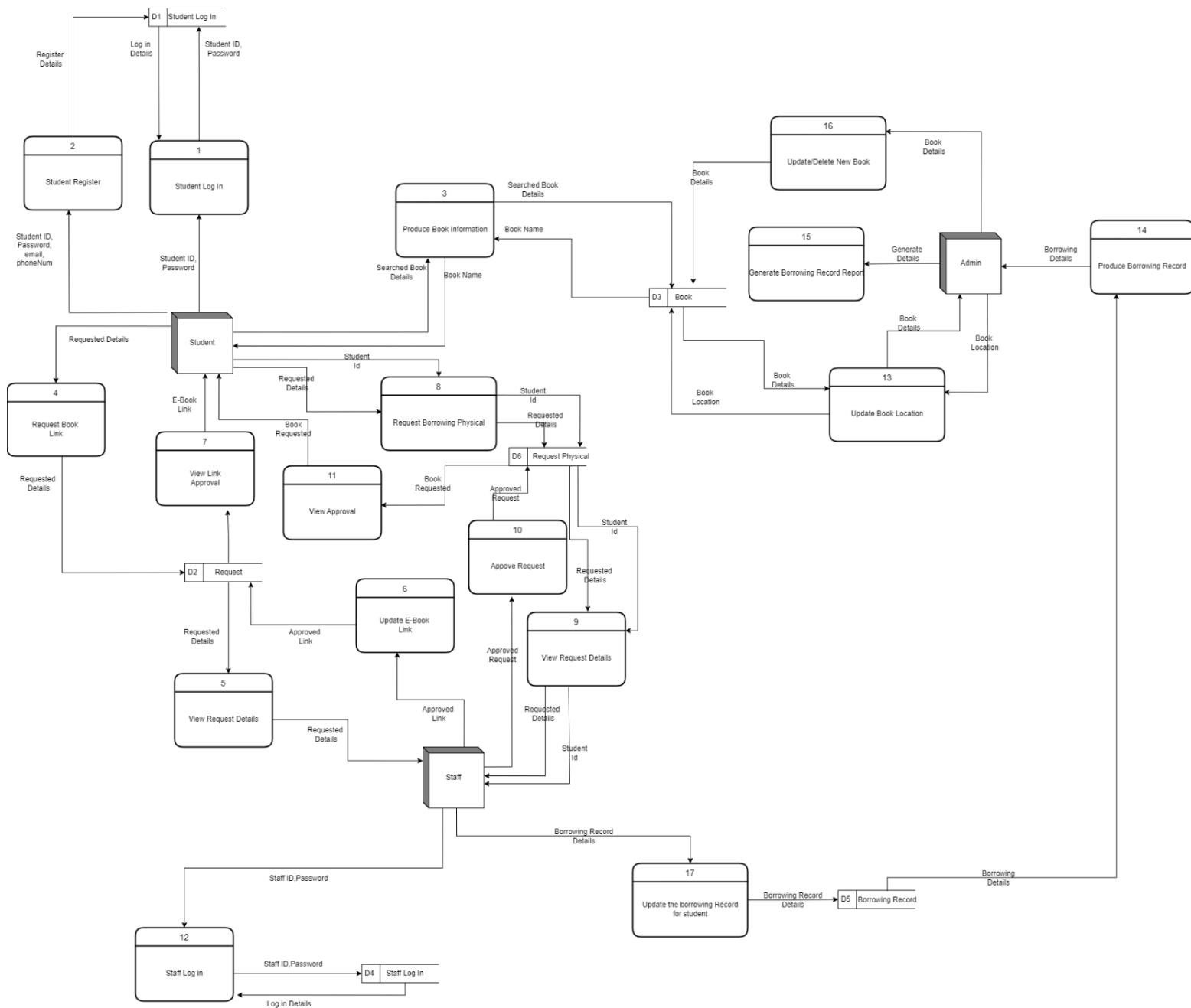
4.3 Data Flow Diagram

4.2.1 Context Diagram



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4.2.2 Diagram o



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5. Conclusion and Recommendation

5.1 Conclusion

The Library Management System (LMS) project analyzes requirements, creates a user-friendly interface, and builds a reliable web-based system with the goal of improving library efficiency and the user experience for administrators, staff, and students. The project guarantees a complete comprehension of user requirements and compliance standards through extensive stakeholder participation, in-depth requirement analysis, and prioritization.

Making an interface that is accessible and easy to use is the main goal of the user-centered design approach, which includes usability testing. Through the implementation of robust security measures, scalable architecture design, and proper technology selection, the project guarantees a dependable and future-proof solution. The system's functionality and user adoption are further supported by thorough testing, thorough integration with current systems, and thorough documentation.

Overall, the project meets its goals by providing an efficient, user-friendly, and secure LMS that dramatically enhances the library experience for all users.

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5.2 Recommendation

To ensure the Library Management System (LMS) remains effective and continues to meet the evolving needs of its users, several recommendations and potential future upgrades should be considered:

1. Enhanced User Experience

- **Voice-Activated Search:** This feature makes the system more accessible to people with impairments by utilizing voice recognition technology to enable users to search for resources using voice commands.
- **Multilingual Support:** To better serve a varied user base and increase accessibility for non-native speakers, include support for multiple languages.
- **Personalization elements:** Put in place elements like user dashboard customization and personalized recommendations to make the user experience more unique.

2. Advanced Integration and Interoperability

- **Mobile Application Integration:** Mobile Application Integration: Create mobile applications that offer easy-to-use mobile access to LMS functions, including notifications, reservations, and book searches.
- **Enhanced API Capabilities:** Expand the API's functionality to enable more complex connections with outside academic and research databases, hence increasing the range of resources that are accessible.

3. Robust Data Analytics and Reporting

- **Advanced Analytics:** To gain a deeper understanding of user behavior, resource usage, and overall system performance, integrate advanced data analytics solutions. This can aid in spotting patterns and allocating resources as efficiently as possible.
- **Predictive Analytics:** Utilize machine learning to predict future trends and user needs, enabling proactive resource management and personalized user experiences.